Day 3: Blocks in the Third Octet (/17 to /23)

Level 1, Month 1, Networking foundations

What I learned

The mask decides which number of the address carries the blocks:

Mask	Number that carries the blocks	Boundary
/9 to /15	Second	16
/17 to /23	Third	24
/25 to /30	Fourth	32

Two different subtractions answer two different questions:

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Bits d'hote                     = 32 - mask
Bits in the number with blocks  = boundary - mask
Taille du bloc                  = 2 ^ (bits in that number)
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Rules for a block in the third octet:

- The third and fourth numbers work together as one counter. They are not independent.
- Adresse reseau: block start, fourth number = 0.
- Premiere adresse: reseau + 1 on the fourth number only.
- Broadcast: third number = (next block start) - 1, fourth number = 255.
- Derniere adresse: broadcast - 1 on the fourth number only.
- The block never starts at the given number. It starts at the largest multiple of the block size that does not exceed it.

Problems solved

Address	Bits d'hote	Taille du bloc	Reseau	Premiere	Broadcast
192.168.20.150/28	4	16	192.168.20.144	192.168.20.145	192.168.20.159

Address	Bits d'hote	Taille du bloc	Reseau	Premiere	Broadcast
172.16.75.20/20	12	16	172.16.64.0	172.16.64.1	172.16.79.255
192.168.100.50/21	11	8	192.168.96.0	192.168.96.1	192.168.103.25
10.20.150.9/18	14	64	10.20.128.0	10.20.128.1	10.20.191.255

Error log

Mistake	Correct answer	Why
172.16.75.20/20: taille du bloc 4	16	4 is the number of bits in the third number. The block size is 2^4.
Block started at 75	Block starts at 64	The block starts at a multiple of 16, not at the given number.
Kept the fourth number as 20	0 in reseau, 255 in broadcast	The fourth number is part of the host counter.
Premiere 172.16.65.0, derniere 172.16.78.0	.64.1 and .79.254	I changed the third number by 1. I must change only the fourth.
192.168.100.50/21: bits d'hote 1, block size 2	11 and 8	I subtracted the wrong numbers. 32 - 21 = 11, and 24 - 21 = 3, then 2^3 = 8.
Broadcast 192.168.101.255	192.168.103.255	Block started at 100 instead of 96, so the end was wrong.
10.20.150.9/18: bits d'hote 6	14	6 is the bits in the third number. Bits d'hote = 32 - 18 = 14.

Main lessons

- I confused two numbers three times: bits d'hote (32 - mask) and bits in the third number (24 - mask). They answer different questions.
- The third and fourth numbers move together. Change only the fourth for first and last host.
- The block start always comes from the table of blocks, never from the given address.

- Check: derniere comes right before broadcast, and the third number stays inside the block.

Anki

Add cards for the three boundaries (16, 24, 32) and for the two subtractions.

Next session

Mixed problems with different masks (/19, /22, /25, /30), then a first look at a pcap file.